

Twisted K-theory

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Abstract

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1 K-theory

1.1 The basics

1.2 Basic definitions in complex K -theory

Throughout, let X be a compact Hausdorff space.

Definition 1.1. Let $\text{Vect}_{\mathbb{C}}(X)$ denote the set of isomorphism classes of finite-rank complex vector bundles on X . It is a semiring under

$$[E] + [F] := [E \oplus F], \quad [E] \cdot [F] := [E \otimes F].$$

The complex K -group of X is the group completion

$$K^0(X) := \text{Groth}(\text{Vect}_{\mathbb{C}}(X)).$$

Thus every class in $K^0(X)$ may be written formally as

$$[E] - [F].$$

Example 1.2. Since every complex vector bundle over a point is trivial, $\text{Vect}_{\mathbb{C}}(*) \cong \mathbb{N}$ with addition given by rank. Therefore

$$K^0(*) \cong \mathbb{Z}.$$

Definition 1.3. Let X_* be a based compact Hausdorff space with basepoint inclusion $i : * \hookrightarrow X_*$. The reduced K -group is

$$\tilde{K}^0(X_*) := \ker(i^* : K^0(X_*) \rightarrow K^0(*)).$$

Equivalently, $\tilde{K}^0(X_*)$ consists of virtual vector bundles of virtual rank 0 at the basepoint.

If X is unbased and $X^+ := X \sqcup \{*\}$ denotes X with a disjoint basepoint, then $K^0(X) \cong \tilde{K}^0(X^+)$.

Proposition 1.4. Two vector bundles E and F define the same element of $K^0(X)$ if and only if there exists a trivial bundle ϵ^n such that

$$E \oplus \epsilon^n \cong F \oplus \epsilon^n.$$

Proposition 1.5. Every class in $K^0(X)$ can be written as

$$[E] - [\epsilon^n]$$

for some vector bundle E over X and some $n \geq 0$.

Proposition 1.6. A vector bundle E represents zero in $\tilde{K}^0(X_*)$ if and only if E is stably isomorphic to a trivial bundle, i.e. if and only if there exist m, n such that

$$E \oplus \epsilon^m \cong \epsilon^n.$$

Definition 1.7. For $n \geq 0$, define the negative reduced K -groups by suspension:

$$\tilde{K}^{-n}(X_*) := \tilde{K}^0(\Sigma^n X_*),$$

where $\Sigma^n X_*$ denotes the n -fold reduced suspension. For an unbased compact Hausdorff space X , define

$$K^{-n}(X) := \tilde{K}^{-n}(X^+) = \tilde{K}^0(\Sigma^n X^+).$$

More generally, for a pair (X, Y) , define

$$K^{-n}(X, Y) := \tilde{K}^0(\Sigma^n(X/Y)).$$

Theorem 1.8 (Bott periodicity). For every compact Hausdorff based space X_* , multiplication by the Bott class induces an isomorphism

$$\tilde{K}^0(X_*) \xrightarrow{\sim} \tilde{K}^0(\Sigma^2 X_*),$$

given by

$$[E] \mapsto ([H] - 1) * [E].$$

where H is the class of the canonical line bundle over $S^2 \cong \mathbb{C}P^1$. Equivalently,

$$\tilde{K}^0(X_*) \cong \tilde{K}^{-2}(X_*).$$

Thus complex K -theory is 2-periodic $K^{-n}(X) \cong K^{-n-2}(X)$. Let $U := \varinjlim U(n)$ and $BU := \varinjlim BU(n)$. Then BU classifies stable complex vector bundles, while U classifies odd K -theory.

Theorem 1.9. For compact Hausdorff X ,

$$\begin{aligned} K^0(X) &\cong [X, \Omega U] \cong [X, \mathbb{Z} \times BU] \\ \tilde{K}^0(X_*) &\cong [X_*, \Omega U]_* \cong [X_*, \mathbb{Z} \times BU]_* \end{aligned}$$

The factor $\mathbb{Z}$ records the virtual rank. Passing to reduced K -theory kills this rank, leaving only BU . In particular, if X is connected, then the basepoint must be sent to the component of BU corresponding to virtual rank 0.

More generally,

$$\tilde{K}^{-n}(X_*) \cong [X_*, \Omega^n(\mathbb{Z} \times BU)]_*.$$

Using Bott periodicity,

$$\Omega^2(\mathbb{Z} \times BU) \simeq \mathbb{Z} \times BU,$$

and also

$$\Omega(\mathbb{Z} \times BU) \simeq U.$$

Therefore

$$\tilde{K}^{-n}(X_*) \cong \begin{cases} [X_*, BU]_* & n \text{ even,} \\ [X_*, U]_* & n \text{ odd.} \end{cases}$$

Theorem 1.10. For a pair (X, Y) , complex K -theory has a long exact sequence

$$\cdots \rightarrow K^n(X, Y) \rightarrow K^n(X) \rightarrow K^n(Y) \rightarrow K^{n+1}(X, Y) \rightarrow \cdots$$

Using the definition

$$K^n(X, Y) := \tilde{K}^n(X/Y),$$

this is the usual long exact sequence associated to the cofiber sequence

$$Y_+ \rightarrow X_+ \rightarrow X/Y.$$

1.3 K-theory of operators

Let V be a vector bundle over X .

Definition 1.11. The compactly supported K -theory of V is defined by

$$K_c^*(V) := \tilde{K}^*(\overline{V}, \partial\overline{V}),$$

where $\overline{V}$ is the fiberwise one-point compactification of V .

In particular, classes of $K_c^0(V)$ can be represented by vector bundles E, F over X together with an isomorphism of π^*E and π^*F away from the zero section, where $\pi : V \rightarrow X$ is the projection.

Let H be a separable infinite-dimensional Hilbert space.

Definition 1.12. Let F be a bounded operator on H . We say F is a *Fredholm operator* if its kernel and cokernel are finite-dimensional. Equivalently, F is Fredholm if there exists a bounded operator G such that $FG - \text{Id}$ and $GF - \text{Id}$ are compact operators. The space of Fredholm operators on H is denoted $\text{Fred}(H)$.

Theorem 1.13 (Atiyah). The space $\text{Fred}(H)$ is homotopy equivalent to $\mathbb{Z} \times BU$. In particular

$$[X, \text{Fred}(H)] \cong K^0(X)$$

1.4 Differential operators and symbols

Let X be a smooth compact manifold and $V, W \rightarrow X$ complex vector bundles. A differential operator of order $\leq k$

$$P : C^\infty(X, V) \rightarrow C^\infty(X, W)$$

is locally

$$P = \sum_{|\alpha| \leq k} a_\alpha(x) \partial_x^\alpha, \quad a_\alpha(x) \in \text{Hom}(V_x, W_x).$$

The principal symbol is

$$\sigma_k(P)(x, \xi) = \sum_{|\alpha|=k} a_\alpha(x) \xi^\alpha \in \text{Hom}(V_x, W_x),$$

which defines a global section

$$\sigma_k(P) \in \Gamma(T^*X, \text{Hom}(\pi^*V, \pi^*W)), \quad \pi : T^*X \rightarrow X.$$

We say P is **elliptic** if $\sigma_k(P)(x, \xi)$ is invertible for all $\xi \neq 0$, i.e. over $T^*X \setminus X$ (the complement of the zero section).

An elliptic P determines a class

$$[\sigma(P)] = [\pi^*V, \pi^*W, \sigma_k(P)] \in K_c^0(T^*X),$$

since $\sigma_k(P) : \pi^*V \rightarrow \pi^*W$ is an isomorphism away from the zero section, which is compact.

Example 1.14 (The exterior derivative is not elliptic). Consider the exterior derivative

$$d : \Omega^k(M) \rightarrow \Omega^{k+1}(M).$$

Its principal symbol is

$$\sigma(d, \xi) = \xi \wedge - : \Lambda^k T_x^* M \rightarrow \Lambda^{k+1} T_x^* M.$$

which we can compute by choosing a smooth function f with $f(x) = 0$ and $df_x = \xi$ and setting $\sigma(d, \xi)(v) = P(f\tilde{v})(x)$ where $\tilde{v}$ is any local extension of v .

This works because the highest-order part of P is exactly what detects the first derivative of f . Lower-order terms vanish because $f(x) = 0$. For $P = d$, take $\omega \in \Lambda^k T_x^* M$, extend it to a local k -form $\tilde{\omega}$, and choose $f \in C^\infty(M)$ with $f(x) = 0$ and $df_x = \xi$. Then $d(f\tilde{\omega}) = df \wedge \tilde{\omega} + f d\tilde{\omega}$.

Evaluating at x , the second term vanishes because $f(x) = 0$, so $d(f\tilde{\omega})(x) = df_x \wedge \omega = \xi \wedge \omega$. Therefore $\sigma(d, \xi)(\omega) = \xi \wedge \omega$. This map is not invertible for $\xi \neq 0$. Indeed, if $\omega = \xi \wedge \eta$, then

$$\sigma(d, \xi)(\omega) = \xi \wedge (\xi \wedge \eta) = 0,$$

so $\sigma(d, \xi)$ has nontrivial kernel. Therefore d is not elliptic.

However, the operator

$$D = d + d^* : \Omega^{\text{even}}(M) \rightarrow \Omega^{\text{odd}}(M)$$

is elliptic. Its symbol is

$$\sigma(D, \xi) = \xi \wedge - - \iota_{\xi^\sharp},$$

and one checks that

$$\sigma(D, \xi)^2 = -|\xi|^2 \text{Id},$$

so $\sigma(D, \xi)$ is invertible for all $\xi \neq 0$. The analytic index is

$$\text{ind}(D) = \dim H_{\text{dR}}^{\text{even}}(X) - \dim H_{\text{dR}}^{\text{odd}}(X) = \chi(X).$$

Example 1.15 (The Dolbeault operator). Let X be a compact complex manifold and $E \rightarrow X$ a holomorphic vector bundle. The Dolbeault operator

$$\bar{\partial}_E + \bar{\partial}_E^* : \Omega^{0, \text{even}}(X, E) \rightarrow \Omega^{0, \text{odd}}(X, E)$$

is elliptic. Its index is

$$\text{ind}(\bar{\partial}_E) = \sum_i (-1)^i \dim H^i(X, E) = \chi(X, E).$$

Example 1.16 (The signature operator). If X is an oriented compact manifold of dimension $4k$, the signature operator is an elliptic operator whose index is $\text{sign}(X)$ the signature of the cup product pairing on $H^{2k}(X; \mathbb{R})$.

1.5 Families of differential operators

Let $Y \rightarrow X \xrightarrow{p} Z$ be a smooth fiber bundle of compact manifolds. Thus for each $z \in Z$, the fiber $X_z := p^{-1}(z)$ is a compact smooth manifold diffeomorphic to Y . Let $V, W \rightarrow X$ be complex vector bundles on the total space X . Their restrictions to the fiber X_z are denoted $V_z := V|_{X_z}$ and $W_z := W|_{X_z}$.

Definition 1.17. A family of differential operators along the fibers is a smoothly varying collection of operators $P_z : C^\infty(X_z, V_z) \rightarrow C^\infty(X_z, W_z)$.

Let $T(X/Z) := \ker(dp) \subset TX$ be the vertical tangent bundle, and let $T^*(X/Z) := T(X/Z)^\vee$ be the vertical cotangent bundle. If $\pi : T^*(X/Z) \rightarrow X$ is the projection, then the principal symbol of P is a bundle map $\sigma(P) : \pi^*V \rightarrow \pi^*W$ over $T^*(X/Z)$.

Locally, if $x_1, \dots, x_n$ are coordinates along the fiber and $z_1, \dots, z_m$ are coordinates on the base, then P has the form

$$P = \sum_{|\alpha| \leq k} a_\alpha(x, z) \partial_x^\alpha$$

There are no derivatives in the z -directions. The principal symbol is

$$\sigma_k(P)(x, z, \xi) = \sum_{|\alpha|=k} a_\alpha(x, z) \xi^\alpha$$

where $\xi \in T_x^*(X_z)$. The family P is called **elliptic** if $\sigma_k(P)(x, z, \xi) : V_{(x,z)} \rightarrow W_{(x,z)}$ is an isomorphism for every nonzero vertical covector $\xi \neq 0$. Equivalently, $\sigma(P)$ is an isomorphism on $T^*(X/Z) \setminus X$, where $X \subset T^*(X/Z)$ is the zero section.

Proposition 1.18. An elliptic family determines a compactly supported K -theory class

$$[\sigma(P)] \in K_c^0(T^*(X/Z))$$

Explicitly, $[\sigma(P)] = [\pi^*V, \pi^*W, \sigma(P)]$.

Example 1.19 (A simple family on circles). Let $X = S^1 \times Z \rightarrow Z$ be the trivial fibration, with fiber coordinate θ on S^1 . Let $V = W = \mathbb{C}$ be the trivial line bundle. Define

$$P_z : C^\infty(S^1) \rightarrow C^\infty(S^1)$$

by

$$P_z = \frac{1}{i} \frac{d}{d\theta} + f(z),$$

where $f : Z \rightarrow \mathbb{R}$ is a smooth function. This is a family of differential operators along the fibers.

The principal symbol only sees the highest-order part, so

$$\sigma(P)(\theta, z, \xi) = \xi : \mathbb{C} \rightarrow \mathbb{C}.$$

For $\xi \neq 0$, multiplication by ξ is invertible. Hence P is elliptic, and it defines a class

$$[\sigma(P)] \in K_c^0(T^*(S^1 \times Z/Z)).$$

Since $T^*(S^1 \times Z/Z) \cong T^*S^1 \times Z$, this is a class in

$$K_c^0(T^*S^1 \times Z).$$

1.6 The analytic index of a family

Theorem 1.20. For each $z \in Z$, ellipticity implies that $P_z : C^\infty(X_z, V_z) \rightarrow C^\infty(X_z, W_z)$ is Fredholm. Hence each P_z has finite-dimensional kernel and cokernel.

If the dimensions of $\ker P_z$ and $\operatorname{coker} P_z$ are locally constant in z , then these form vector bundles $\ker P \rightarrow Z$ and $\operatorname{coker} P \rightarrow Z$, and one defines $\operatorname{Ind}_{\text{an}}(P) := [\ker P] - [\operatorname{coker} P] \in K^0(Z)$.

In general, the dimensions may jump, so $\ker P$ and $\operatorname{coker} P$ need not be vector bundles.

Example 1.21 (Jumping kernels in a family). Consider the family of elliptic operators on S^1

$$P_t = \frac{1}{i} \frac{d}{d\theta} + t,$$

where $t \in \mathbb{R}$. Acting on Fourier modes, we have

$$P_t(e^{in\theta}) = (n + t)e^{in\theta}.$$

Therefore P_t has kernel precisely when $n + t = 0$ for some $n \in \mathbb{Z}$. Equivalently,

$$\dim \ker P_t = \begin{cases} 1 & t \in \mathbb{Z}, \\ 0 & t \notin \mathbb{Z}. \end{cases}$$

Thus the dimensions of the kernels jump as t varies. Consequently the kernels do not form a vector bundle over the parameter space.

For a smooth family of elliptic operators $P_z : C^\infty(X_z, V_z) \rightarrow C^\infty(X_z, W_z)$, choose Sobolev completions. For $s \gg 0$, each P_z extends to a bounded Fredholm operator

$$P_z : H^s(X_z, V_z) \rightarrow H^{s-k}(X_z, W_z)$$

where $H^s(X_z, V_z)$ = sections of V_z with s weak derivatives in L^2 . As z varies, the spaces $H^s(X_z, V_z)$ and $H^{s-k}(X_z, W_z)$ form Hilbert bundles over Z . In fact being infinite dimensional, the fibers are all abstractly modeled on the same Hilbert space.

The family P is therefore a bundle map between Hilbert bundles

$$\mathcal{H}_V^s \longrightarrow \mathcal{H}_W^{s-k}.$$

Theorem 1.22 (Kuiper's theorem). Let H be an infinite-dimensional separable Hilbert space. Then the unitary group $U(H)$, equipped with the norm topology, is contractible.

Corollary 1.23. Let Z be paracompact, and let $\mathcal{H} \rightarrow Z$ be a Hilbert bundle whose fibers are infinite-dimensional separable Hilbert spaces and whose structure group is $U(H)$ with the norm topology. Then $\mathcal{H}$ is globally trivial:

$$\mathcal{H} \cong Z \times H.$$

Therefore, after choosing trivializations of the Hilbert bundles, the family P becomes a continuous map

$$Z \rightarrow \text{Fred}(H).$$

By Atiyah's theorem, the homotopy class of this map defines a class in

$$[Z, \text{Fred}(H)] \cong K^0(Z).$$

The corresponding class is called the analytic index of the family and is denoted

$$\text{Ind}_{\text{an}}(P) \in K^0(Z).$$

1.7 The Thom isomorphism in K theory

Let X be a finite CW complex, and let $E \rightarrow X$ be a real vector bundle of rank n . Let $D(E)$ and $S(E)$ denote its disk and sphere bundles. The Thom space of E is

$$\text{Th}(E) := D(E)/S(E).$$

Equivalently,

$$K_c^*(E) := K^*(D(E), S(E)) \cong \tilde{K}^*(\text{Th}(E)).$$

For each $x \in X$, the fiber pair is $(D(E_x), S(E_x)) \cong (D^n, S^{n-1})$. Therefore

$$K^*(D(E_x), S(E_x)) \cong \tilde{K}^*(S^n).$$

By Bott periodicity,

$$\tilde{K}^i(S^n) \cong \begin{cases} \mathbb{Z} & i \equiv n \pmod{2}, \\ 0 & i \not\equiv n \pmod{2}. \end{cases}$$

A generator of this copy of $\mathbb{Z}$ is called the Bott generator of the fiber.

Definition 1.24 (K -orientation). A K -orientation of $E \rightarrow X$ is a class

$$u_E \in K^n(D(E), S(E))$$

whose restriction to every fiber pair $(D(E_x), S(E_x))$ is the Bott generator

$$u_E|_{(D(E_x), S(E_x))} \in K^n(D(E_x), S(E_x)) \cong \mathbb{Z}.$$

The class u_E is called the K -theoretic Thom class of E .

Here K^n is understood modulo Bott periodicity, so only the parity of n matters.

Theorem 1.25 (Thom isomorphism in complex K -theory). Let $E \rightarrow X$ be a K -oriented real vector bundle of rank n , with Thom class

$$u_E \in K^n(D(E), S(E)).$$

Then cupping with u_E gives an isomorphism

$$K^i(X) \xrightarrow{\sim} K^{i+n}(D(E), S(E)), \quad \alpha \mapsto \pi^* \alpha \smile u_E.$$

Equivalently,

$$K^i(X) \xrightarrow{\sim} K_c^{i+n}(E).$$

In particular, every complex vector bundle is canonically K -oriented, because a complex vector bundle has a canonical Spin^c -structure.

1.8 The topological index

Let $p : X \rightarrow Z$ be a smooth fiber bundle with compact fibers. Let $T(X/Z) := \ker(dp)$ and $T^*(X/Z) := T(X/Z)^\vee$. An elliptic family P gives a symbol class $[\sigma(P)] \in K_c^0(T^*(X/Z))$. The topological index is a pushforward map $\text{Ind}_{\text{top}} : K_c^0(T^*(X/Z)) \rightarrow K^0(Z)$.

Choose a fiberwise embedding $i : X \hookrightarrow Z \times \mathbb{R}^N$, and let $\nu \rightarrow X$ be its normal bundle. Since i is an embedding over Z , there is an isomorphism $T(X/Z) \oplus \nu \cong X \times \mathbb{R}^N$. This induces, after passing to cotangent directions, an embedding $T^*(X/Z) \hookrightarrow Z \times \mathbb{R}^{2N}$. The normal bundle of this embedding has a canonical complex structure after stabilization, so the embedding is K -oriented.

Theorem 1.26 (Pushforward for a K -oriented embedding). Let $j : M \hookrightarrow N$ be a smooth embedding, and let $\nu \rightarrow M$ be its normal bundle. Suppose that ν is equipped with a complex structure, or more generally a K -orientation. Then there is a pushforward map $j_! : K_c^0(M) \rightarrow K_c^0(N)$.

It is defined as follows. Choose a tubular neighborhood $U \subset N$ of M , so that $U \cong \nu$. The Thom isomorphism for the complex vector bundle $\nu \rightarrow M$ gives an isomorphism $K_c^0(M) \xrightarrow{\sim} K_c^0(\nu)$. Under the identification $U \cong \nu$, this gives a class in $K_c^0(U)$. Extending by zero outside $U \subset N$

gives a class in $K_c^0(N)$. Thus $j_!$ is the composite $K_c^0(M) \xrightarrow{\text{Thom}} K_c^0(\nu) \cong K_c^0(U) \rightarrow K_c^0(N)$. Finally, Bott periodicity identifies $K_c^0(Z \times \mathbb{R}^{2N}) \cong K^0(Z)$. The topological index is the composite

$$K_c^0(T^*(X/Z)) \xrightarrow{i_!} K_c^0(Z \times \mathbb{R}^{2N}) \cong K^0(Z).$$

Theorem 1.27 (Atiyah-Singer index theorem for families). The analytic index and the topological index coincide:

$$\text{Ind}_{\text{an}}(P) = \text{Ind}_{\text{top}}([\sigma(P)]) \in K^0(Z).$$

1.9 Clifford algebras